

ZipCode3.0 - Test Document

Basic Functional Test Cases

Test Case	Input	Expected Output
1. Search for existing ZIP code	-z10001	Full record printed for 10001
2. Search for multiple valid ZIPs	-z10001 -z90210 -z33101	3 valid records displayed
3. Search for non-existent ZIP	-z99999	"Zip Code 99999 not found in the file."
4. Mix of valid and invalid	-z10001 -z99999	1 record printed, 1 not found
5. No ZIP codes provided	none, just filename	Program enters interactive mode
6. ZIP flag but no file	-z10001	Error: Please specify the filename before zip code flags

Edge Case Test Cases

Test Case	Input	Expected Output
7. ZIP code with letters	-zABCDE	Error: invalid zip code argument
8. Negative ZIP code	-z-12345	Error: invalid zip code argument
9. Very large number	-z999999999	Likely not found or overflow behavior
10. Duplicate ZIP codes	-z10001 -z10001	Same record printed twice
11. ZIP with leading zero	-z00501	Correct record shown if present (leading 0 dropped)

Notes

- All tests should be performed on a structured, length-indicated file generated from the ConvertToBlocked step.
- The index file must also be built with BuildIndex before using search-related functionality.

- Each test should be validated by comparing terminal output to expected messages and content.

Tools Required

- Terminal or shell for command-line interaction
- myProgram, BuildIndex, and ConvertToBlocked executables compiled
- Valid input files: zipCodes.txt, zipCodes.bss, zipCodes.idx